

AB RENEWABLE ENERGY ENGINEERING

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SOLAR THERMAL

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1. The average power density of solar heat striking the earth surface.

- a. 1 kW/m²
- ☒ b. 0.89 kW/m²
- c. 0.58 kW/m²
- d. None of the above

$$\text{Solar constant} = 1.35 \frac{\text{kW}}{\text{m}^2}$$

2. The factor affecting solar radiation.

- a. Cloudiness
- b. Overcasting
- ☒ c. Shade
- d. All of the above

3. The place located near the equator will receive ____ solar radiation than those which are far from it.

- a. less
- b. the same
- ☒ c. more
- d. None of the above

4. The basis for determining the degree of inclination of the solar collector.

- ☒ a. Latitude of the place
- b. Longitude of the place
- c. Elevation of the place
- d. None of the above

5. The device used to convert solar radiation into heat.

- ☒ a. Solar absorber
- b. Solar cell
- c. Solar panel
- d. None of the above

6. A type of solar collector that follows the movement of the sun.

- a. Flat-plate-type collector
- ☒ b. Concentrating-type collector
- c. Air-inflated-type collector
- d. None of the above

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7. Determine the size of the flat-plate solar collector perpendicularly inclined at 11 deg to the direction of the sunlight which is required to supply thermal heat to a fruit dryer that consumes 1000 kcal per hour of heat energy. Assume a 0.89 kW per m² average power density and a 0.77 transmission factor.

- a. 10.32 m²
- b. 12.65 m²
- ☒ c. 14.15 m²
- d. None of the above

8. Non-tract-type solar collector.

- ☒ a. Flat-plate-type collector
- b. Concentrating-type collector
- c. Air-inflated-type collector
- d. None of the above

9. A solar collector purposely to reflect back the heat of a given space to increase its temperature.

- a. Flat-plate-type collector
- ☒ b. Concentrating-type collector
- c. Air-inflated-type collector
- d. None of the above

10. Collection efficiency of a high-temperature system solar collector.

- a. Up to 60%
- b. Up to 70%
- ☒ c. Up to 80%
- d. None of the above

11. Collection efficiency of a low-temperature system solar collector.

- a. Up to 60%
- ☒ b. Up to 70%
- c. Up to 80%
- d. None of the above

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12. Parabolic solar collector is an example of ____.

- a. non-concentrating type SC
- ☒ b. concentrating type SC
- c. non-tract type solar collector
- d. None of the above

13. Suitable heat collector for solar cell.

- a. High-temperature flat-plate collector
- b. Low-temperature parabolic solar collector
- ☒ c. Air-inflated type collector
- d. None of the above

14. What is the collection efficiency of a solar cooker that delivers a 1kW thermal energy to a cooking pot if it has a 1.2 m² collector area? The average power density is 0.89 kW/m².

- a. 81.7 %
- b. 89.3 %
- ☒ c. 93.6 %
- d. None of the above

15. The solar tracker in a solar power system is ____.

- a. used to carry the solar collector in areas where sun energy is most efficient
- ☒ b. used to tilt the solar collector to follow the direct position of the sun
- c. used to monitor the position of the sun
- d. None of the above

16. Which of the following statements is true?

- a. The intensity of solar radiation increases as the distance of the location from the equator increases.
- ☒ b. The intensity of solar radiation decreases as the distance of the location from the equator increases.
- c. The intensity of solar radiation does not change with the distance of the location from the equator.
- d. None of the above

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17. Two hundred liters of water is to be heated from 30°C to 80°C in 1 hour using solar heat. What is the power required to heat the given amount of water?

- a. 9 kW
- b. 11 kW
- ☒ c. 12 kW
- d. None of the above

18. What is the area of the tube-type solar collector required for the heater in heating the 200 liters of water in Item 28 above, if the collection efficiency is 80% and is oriented direct to the sunlight. The average power density of the sun is 0.89 kW/m².

- a. 14.3 m²
- ☒ b. 16.8 m²
- c. 19.6 m²
- d. None of the above

19. Concentrating reflector ____ the radiation intensity of the absorbing surface.

- ☒ a. increases
- b. decreases
- c. maintains
- d. None of the above

20. Which of the following materials provide high reflection value?

- a. Mirror glass
- b. Aluminum foil
- ☒ c. Electroplated silver
- d. None of the above

21. The solar collector's heat absorbing capacity can be optimized by ____.

- a. increasing its insulation thickness
- b. providing the solar collector surface with glass mirror
- ☒ c. painting the solar collector with dull black color
- d. None of the above

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GOD BLESS !!!

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